LinkedIn edunet foundation

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Artificial Intelligence in the Contemporary World

Project Domain : Data Literacy and Responsible AI

Title : Semester Smart – Structured Planning for Teachers, Instant Answers for Students

[Excel / AI Chatbot/ Copilot]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

Faculty Name
NAMRATA GAUR

Faculty Email
namo1999d@gmail.com

Guide
Umamaheshwari R

Institution:

Independent Researcher (Former JPF, NCERT)

Year - 2026

3. Problem Statement

In higher education institutions, academic planning and communication rely heavily on scattered and static data sources such as spreadsheets, academic calendars, notices, and informal classroom communication. Faculty members manage multiple responsibilities including scheduling internal assessments, assignments, examinations, and balancing these with research and professional development activities. However, the absence of a structured and integrated system makes it difficult to visualize the complete semester timeline and distribute workload effectively. As a result, academic activities are often unevenly spread, leading to periods of excessive workload followed by underutilized time.

At the same time, students, particularly those in their first year, face significant confusion in understanding academic schedules, important deadlines, and overall semester structure. Although information is technically available, it is not presented in a consolidated, interactive, or easily accessible format. This lack of clarity leads to missed deadlines, poor planning, and reduced academic engagement. Students are also unable to see a complete picture of their academic journey within a semester, which limits their ability to manage time effectively and take advantage of opportunities such as self-paced learning or additional courses.

This problem is significant because it directly impacts both teaching efficiency and student learning experience. Inefficient academic planning affects faculty productivity, leaving limited time for research and professional growth. For students, especially new entrants, the lack of structured guidance creates unnecessary stress and confusion, ultimately affecting their academic performance and overall engagement with the learning process.

The primary stakeholders affected by this problem include higher education faculty members, administrative staff, and students. Faculty members require better tools to plan and balance academic and research responsibilities, while students need a simple and accessible way to understand and navigate their academic schedules. Institutions as a whole are also impacted, as inefficient planning and communication reduce overall academic effectiveness.

Existing approaches, such as institutional calendars, spreadsheets, and learning management systems, provide only partial solutions. These tools are typically static, fragmented, and not designed for integrated visualization or interactive use. They do not offer insights into workload distribution, nor do they provide real-time, user-friendly access to information. Additionally, while artificial intelligence is increasingly being used in education, its application in this context is often either too complex or not directly aligned with everyday academic needs. There is a lack of simple, transparent, and responsible AI solutions that can bridge the gap between data availability and meaningful usage.

Therefore, there is a clear need for a system that can transform scattered academic data into a structured, visual, and accessible format, while also enabling instant and user-friendly interaction through an AI assistant. This project addresses this gap by developing a data-driven academic planning and information system that enhances clarity, improves planning efficiency, and supports better engagement for both faculty and students.

4. Objectives

- To collect and structure semester-related academic data including IA schedules, The primary objective of this project is to develop a structured and technology-integrated system for academic planning and information access in higher education. The specific objectives of the study are as follows:
- To collect and understand the structure of academic data related to semester schedules, including internal assessments (IA), assignments, examinations, faculty workload, and institutional calendars.
- To clean and preprocess the raw dataset by organizing it into a structured format, ensuring consistency in dates, removing redundancies, and standardizing academic events for accurate analysis.
- To analyze using tools such as Excel to map the semester timeline, analyze workload distribution, and identify patterns across weeks and months.
- To build interactive dashboards and visual reports that effectively present key academic indicators such as total assessments, total examinations, and faculty workload, enabling better planning and decision-making.
- To design and implement a rule-based AI chatbot that provides instant, user-friendly access to academic information, allowing students to quickly understand schedules, deadlines, and semester structure.
- To leverage AI tools such as Copilot and chatbot systems to enhance data structuring, automate insights generation, and improve accessibility of academic information in a responsible and transparent manner.
- **To develop a comprehensive "Semester Smart" system that integrates academic data, visualization, and AI assistance to streamline semester planning for faculty and provide clear, instant academic guidance to students.**

5. Methodology / Implementation Plan

#	Phase	Description
1	**Data Collection & Exploration**	A structured dataset created based on the academic semester plan, including details such as subjects (Indian Philosophy, Greek Philosophy, Logic, Introduction to Ethics), unit-wise distribution, internal assessments (IA), assignments, self-paced learning schedules, holidays, and break periods. The dataset covers the complete semester timeline from August to December 2025. Initial exploration will involve understanding the distribution of events across months, identifying overlaps between assessments, and ensuring balanced workload allocation. Key variables such

		as event type, subject, and date were analyzed to support effective planning and visualization.
2	**Data Cleaning & Preprocessing**	Standardizing date formats across all entries Categorizing events into defined types (IA, Assignment, Self-paced, Break, Exam) Removing inconsistencies in naming conventions Ensuring no overlapping of major assessments (minimum gap constraints applied) Organizing data into a structured format suitable for pivot tables and dashboard creation.
3	**Analysis & AI Integration**	Data analysis will be performed using Excel functions and pivot tables to derive insights such as: Month-wise distribution of academic events Subject-wise workload analysis Identification of peak workload periods Gap analysis between assessments Functions such as COUNTIF, SUMIF, and pivot aggregations were used to calculate event frequencies and workload distribution. An AI chatbot was developed using Microsoft Copilot Studio to enable students to query academic information. The chatbot provides responses to queries related to IA schedules, assignment deadlines, holidays, self-paced courses, and research-based course recommendations.
4	**Visualization / Output**	An interactive Excel dashboard was developed to visually represent the semester plan. Key features include: KPI indicators (total IA, assignments, self-paced activities, breaks)

		Pivot charts showing event distribution Slicers for filtering data by month Structured layout for easy interpretation of workload Additionally, a rule-based AI assistant was implemented to provide instant responses to student queries such as schedules and holidays, topics and self paced courses related to research interests and improving accessibility of any academic information for the semester
5	**Documentation & Submission**	The entire project workflow including dataset preparation, analysis steps, dashboard design, and AI prompts will be documented. All project files will be uploaded to a GitHub repository, and a demonstration video explaining the working system will be recorded and submitted.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Microsoft Copilot — via Microsoft 365 subscription — where applicable
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

The dataset used in this project is a **structured academic planning dataset** designed to represent the complete semester schedule. It contains information related to four subjects:

Indian Philosophy, Greek Philosophy, Logic, and Introduction to Ethics, along with their respective units, internal assessments (IA), assignments, self-paced learning activities, holidays, and break periods.

The dataset spans the full semester duration from **August 2026 to December 2026**, with **day-wise entries** to ensure detailed planning and tracking. Each record represents a specific academic event mapped to a particular date, subject, and event type.

Key attributes in the dataset include:

Date – Represents each day of the semester

Month – Categorizes data for monthly analysis

Subject – Indicates the academic subject

Event Type – Defines the type of activity (IA, Assignment, Self-paced, Break, Exam)

Description – Provides additional context such as unit coverage or holiday details

The dataset is primarily **categorical and time-based**, enabling analysis of workload distribution, event frequency, and scheduling gaps. It has been manually structured to simulate a real academic semester plan, ensuring that it supports both visualization (dashboard creation) and query-based interaction (chatbot responses).

Column Name	Data Type	Description	Example Value
Date	Date	Represents the specific day of the semester	01-08-2026
Month	Text	Month corresponding to the date, used for filtering and analysis	August
Subject	Text	Name of the subject for which the event is scheduled	Indian Philosophy
Event_Type	Categorical	Type of academic activity	IA / Assignment / Self-paced
Description	Text	Additional details about the event (unit	Unit 1 Coverage / Sunday Break

		info, holiday, break reason, etc.)	
Unit	Text	Indicates the unit being taught in that period	Unit 1
Day_Type	Categorical	Specifies whether it is a working day or break	Working Day / Holiday / Sunday
Workload	Numeric	Represents intensity of the day (used for analysis and visualization)	0 / 50

C. Project Timeline

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

•A well-structured and comprehensive semester dataset covering all academic activities including internal assessments, assignments, self-paced learning, holidays, and break periods across all subjects

An interactive Excel dashboard that provides a consolidated and visually clear view of the semester through pivot tables, charts, and slicers

Improved academic planning by ensuring balanced workload distribution and maintaining appropriate gaps between assessments and assignments

Clear visualization of key performance indicators such as total number of assessments, assignments, self-paced activities, and overall workload intensity

A functional AI chatbot developed using Microsoft Copilot Studio capable of answering student queries related to schedules, holidays, and learning activities

Enhanced accessibility of academic information, enabling students to quickly retrieve relevant details through simple queries instead of manual searching

Better decision-making support for teachers through structured data representation and easy identification of peak workload periods

Promotion of academic data literacy by demonstrating how structured data can be used for planning, analysis, and visualization

A scalable and adaptable framework that can be reused for future semesters or extended to other academic domains

8. References

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning
- Microsoft Corporation. Microsoft Copilot. Available at: https://copilot.microsoft.com
- https://www.du.ac.in/uploads/new-web/09072025_holiday.pdf